REVIEW

of the foreign scientific consultant for the dissertation work of Yesmukhamedov Nurmaganbet Seitkaliuly «Computer System for Processing and Analysis of Eye Data to Support Laser Coagulation in the Treatment of Diabetic Retinopathy», submitted for the degree of Doctor of Philosophy (PhD) on educational program: 8D06102 — Computer and Software Engineering

Relevance of the Research Topic

Diabetic retinopathy remains one of the leading causes of preventable blindness among the working-age population worldwide, and its early stages are clinically asymptomatic. Timely detection therefore depends on the regular screening of colour fundus images, yet the number of ophthalmologists is insufficient to examine manually the rapidly growing population of diabetic patients. This shortage is especially acute in resource-limited and rural healthcare settings, including the regions of Kazakhstan, which makes automated and computationally efficient diagnostic tools a pressing socio-economic need.

The relevance of this research, however, is not confined to its clinical motivation. The dominant tradition in automated diabetic retinopathy diagnosis treats image preprocessing as ancillary data preparation that requires no methodological discussion, on the assumption that a sufficiently deep convolutional neural network learns the necessary invariances directly from raw or minimally normalised pixels. The candidate demonstrates convincingly the limitation of this assumption: fundus images acquired by different cameras, under different illumination conditions and with different noise levels exhibit substantial domain variability, which manifests as distribution shift in the input feature space and degrades the generalisation of models trained on one domain and applied to another. The alternative position advanced in the dissertation — that preprocessing is an integral component of the diagnostic model, because it defines the feature space available to the network and therefore co-determines what the network can learn — is clearly articulated and scientifically well founded.

Purpose of the Dissertation Research and Scientific Results

The aim of the research is to develop and experimentally validate an integrated fundus image enhancement and CNN-based classification framework for automated multi-stage diabetic retinopathy diagnosis, in which an eight-stage preprocessing pipeline is specified as a component of the model rather than as preparation of the data, and to establish what difference that specification makes to diagnostic performance, transferability and interpretability under constrained computational conditions. The objectives proceed systematically from the analysis of the problem domain, through the theoretical foundations of image enhancement and deep learning, the formalisation of the integrated methodology, the experimental programme and the validation of its reliability, to the design of a screening system architecture suitable for resource-limited environments. The structure of the dissertation follows this progression closely, and each chapter develops from the one preceding it.

Scientific Results

The eight-stage integrated preprocessing pipeline is formalised as a binding part of the model specification, comprising canonical flip, optic-disc–fovea rotation normalisation, field-of-view crop with isotropic resize and centred zero-padding, field-of-view mask generation as a fourth input channel, adaptive flat-field illumination correction whose Gaussian parameter scales with the per-image field-of-view diameter, dual-constraint stochastic CLAHE on the LAB luminance channel, augmentation, and dataset-specific channel-wise normalisation. Rotation normalisation relies on a pre-trained and frozen heatmap-regression detector that is not co-trained with the classifier, so that the preprocessing remains a fixed transform throughout.

The effect of this specification was placed under a controlled two-by-two factorial design on the EyePACS corpus across two established architecture families, ResNet-50 and EfficientNet-B3, against an equivalent baseline configuration. The integrated configuration dominated the baseline on both architectures, satisfying a dominance criterion fixed in advance in all three of its components, with the effect surviving correction for multiple comparisons and showing no interaction with the choice of architecture. A cumulative component ablation quantified the contribution of the individual stages and identified the two photometric stages as the leading contributors, while parameter sweeps established interior optima for the CLAHE clip limit and the flat-field parameter, confirmed on held-out data and supported by image quality metrics.

The generalisation properties of the configuration were examined on seven further corpora. A model trained on EyePACS transferred to APTOS 2019 without retraining and cleared the generalisation threshold fixed in advance while exceeding the baseline on every grade; on both external clinical corpora, IDRiD and Messidor-2, the integrated configuration attained higher weighted F1 by a clinically meaningful margin; and classification performance was maintained across all five camera groupings, with the dispersion of performance between those groupings substantially reduced. Grad-CAM analysis, evaluated against pixel-level lesion annotations by means of the Attention–Lesion Overlap metric proposed in the dissertation and by Intersection-over-Union, found model attention more closely aligned with expert annotation on all four annotated lesion types.

Novelty and Contribution

The principal contribution is the conceptual and operational reframing of preprocessing as an integral model component, together with the placement of that reframing under direct and controlled experimental test. Within this framework the candidate contributes a dual-constraint stochastic CLAHE variant adapted to the five-class grading task, which serves at once as an enhancement operator and as a source of regularisation; an adaptive flat-field correction that scales with the per-image field of view and is applied only inside the mask; an explicit binary field-of-view mask introduced as a dedicated stage and appended as a fourth input channel; and the Attention–Lesion Overlap metric, an asymmetric measure of the fraction of a lesion covered by model attention.

I consider one further contribution to be of particular methodological value. The mechanism by which preprocessing is supposed to improve generalisation is habitually invoked in the literature as an explanation but rarely measured. The candidate measures it directly, computing the distance between the training distribution and each external distribution at the penultimate layer under both configurations across six external corpora, at the cost of forward passes only, and under the protocol condition that normalisation uses source-domain statistics and is never recomputed on the target. This converts an explanatory assumption into a falsifiable quantity, and the distance was found to decrease on every corpus examined. Equally noteworthy is the treatment of initialisation: the choice among candidate pretraining strategies was settled by a frozen-backbone linear-probe acceptance gate rather than by assumption, and the candidate reports the negative outcome — that label-free self-supervision trained from scratch failed that gate — as part of the evidence.

Degree of Reliability and Validity of the Scientific Results

The reliability of the results is assured by the scale and diversity of the empirical basis and by the rigour of the statistical apparatus. The work draws on eight datasets organised into functional tiers and spanning four camera manufacturers: EyePACS with approximately 35,126 graded images for primary training and ablation, whose second and unlabelled split supplies the in-domain pretraining corpus; APTOS 2019 for cross-dataset transfer; IDRiD and Messidor-2 for external clinical evaluation and explainability; DDR, ODIR-5K and RFMiD for device domain shift; and a Kazakh clinical dataset for training in a data-scarce regime.

All experiments were conducted under five-fold patient-level stratified cross-validation with strict leakage control, with primary metrics reported as mean and standard deviation across a multi-metric assessment comprising weighted F1, ROC-AUC, Cohen's Kappa with quadratic weights and accuracy. Statistical reliability rests on McNemar and DeLong tests, mixed-effects modelling across folds, bootstrap confidence intervals and Bonferroni/Holm correction for multiple comparisons. I regard it as a particular strength of this work that the success criterion for each hypothesis was fixed before the experiment that tested it, and that the boundary conditions and limitations of the approach are stated explicitly rather than left implicit. This methodology meets international standards for experimental validation in medical image analysis.

Practical Significance of the Results

The integrated pipeline constitutes a fully specified and reproducible preprocessing regime for automated diabetic retinopathy diagnosis under constrained computational conditions. It is defined stage by stage, with its operators, parameters and order of application, and its implementation is reproduced in an appendix, which makes it directly usable by other researchers and suitable for screening environments where high-end hardware is unavailable.

The candidate further proposes a modular architecture for an automated screening system, comprising a configurable preprocessing engine, an inference module, store-and-forward telemedicine support and a physician-in-the-loop decision-support interface, designed for integration with PACS and EHR systems and with the national eHealth platform, and oriented towards screening in rural and underserved regions, including the healthcare infrastructure of Kazakhstan. The system is conceived throughout as decision support in which the clinician retains the diagnosis and the responsibility for it. The approach is scalable and cost-effective, and its data protection provisions are aligned with international requirements.

Academic Contribution and Doctoral Qualification

Throughout our collaboration the doctoral candidate has demonstrated notable growth as a researcher, developing into a mature specialist capable of formulating a research problem independently, of selecting and adapting appropriate methods, and of carrying a complex experimental programme through to completion. He has shown a firm command of modern deep learning methods, of image processing theory and of the apparatus of statistical validation, and he has maintained a standard of scientific honesty in reporting that deserves separate mention. The results of the research have been published in peer-reviewed journals and presented at international conferences, including the 3rd International Workshop on Digital Society in Istanbul, which confirms their recognition by the scientific community.

The dissertation of Yesmukhamedov Nurmaganbet Seitkaliuly makes a significant, original and scientifically substantiated contribution to the field of automated diabetic retinopathy diagnosis. The work meets the academic and methodological requirements for doctoral research and conforms to international standards for the award of a PhD degree. Accordingly, this dissertation is recommended for the awarding of the degree of Doctor of Philosophy (PhD) in the educational program 8D06102 — Computer and Software Engineering.

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